DEPARTMENT OF SOFTWARE ENGINEERING

CLOUD COMPUTING | COURSE CODE: CC

ASSIGNMENT 01

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Assignment Number & Lab Title	Assign 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation
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Course Code	CC
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Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

```
nidaawajid@ubuntu:~/Gitea$ docker compose up -d
[+] up 25/25
 ✔ Image postgres:17-alpine Pulled                                                                                1220.1
 ✔ Image gitea/gitea:latest Pulled                                                                                 623.5
 ✔ Network webnet           Created                                                                                  0.4
 ✔ Volume gitea_postgres    Created                                                                                  0.1
 ✔ Volume gitea             Created                                                                                  0.0
 ✔ Container gitea          Started                                                                                  4.2
 ✔ Container gitea_db       Started                                                                                  3.0
nidaawajid@ubuntu:~/Gitea$ docker compose ps
NAME       IMAGE              COMMAND                  SERVICE   CREATED              STATUS              PORTS
gitea      gitea/gitea:latest   "/usr/bin/entrypoint…"   gitea     About a minute ago   Up 58 seconds       22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db   postgres:17-alpine   "docker-entrypoint.s…"   db        About a minute ago   Up About a minute   5432/tcp
nidaawajid@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
```

Screenshot Name: `task1_gitea_running.png`

Figure 1: Gitea and PostgreSQL containers running on the Ubuntu server with a successful HTTP status from port 3000.

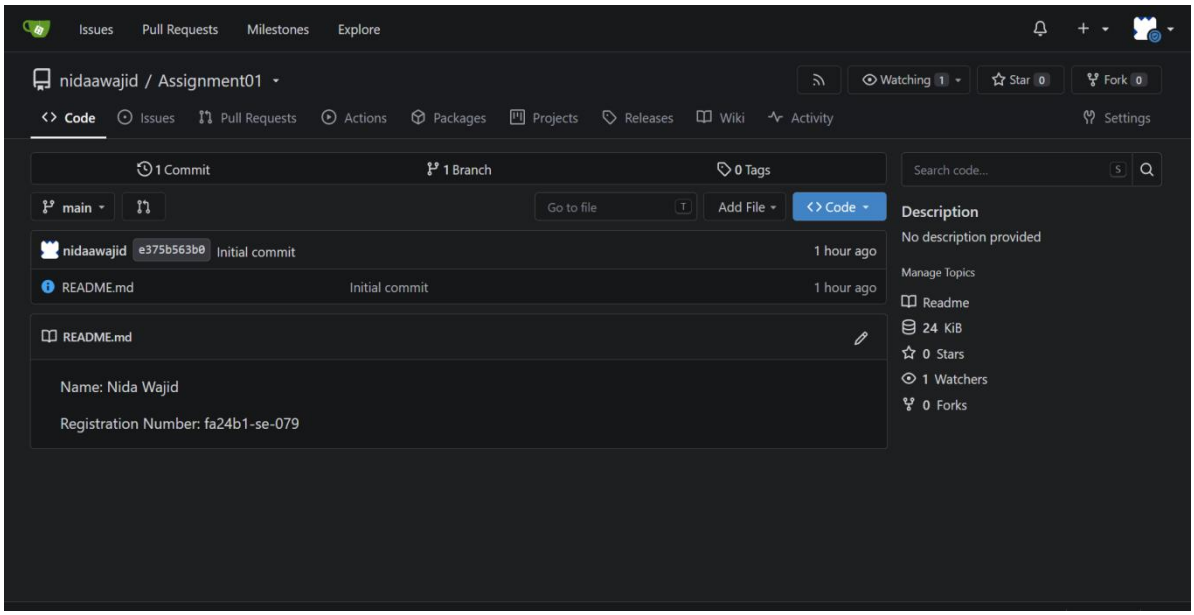
Step 1.2: Create and Push the Initial Repository

```
nidaawajid@ubuntu:~/Assignment01$ git remote -v
gitea   http://nidaawajid@127.0.0.1:3000/nidaawajid/Assignment01.git (fetch)
gitea   http://nidaawajid@127.0.0.1:3000/nidaawajid/Assignment01.git (push)
nidaawajid@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://nidaawajid@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/nidaawajid/Assignment01.git/'
nidaawajid@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://nidaawajid@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 270 bytes | 270.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/nidaawajid/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
nidaawajid@ubuntu:~/Assignment01$ _
```

Screenshot Name: `task1_gitea_push.png`

Figure 2: Initial Assignment01 commit pushed successfully to the local Gitea remote with no token visible.

Step 1.3: Verify the Gitea Repository



Screenshot Name: `task1_gitea_repository.png`

Figure 3: Public Assignment01 repository in Gitea showing the owner, repository name, and rendered README.md.

> **Task 1 result:** *Gitea was installed and run on the Ubuntu server, and a separate local repository was pushed to it over the local loopback interface.*

Task 2: Push the Same Repository to GitHub

Step 2.1: Create the GitHub Repository and Verify Both Remotes



Screenshot Name: `task2_remotes.png`

Figure 4: Output of git remote -v showing both the gitea and github remotes without any token or password.

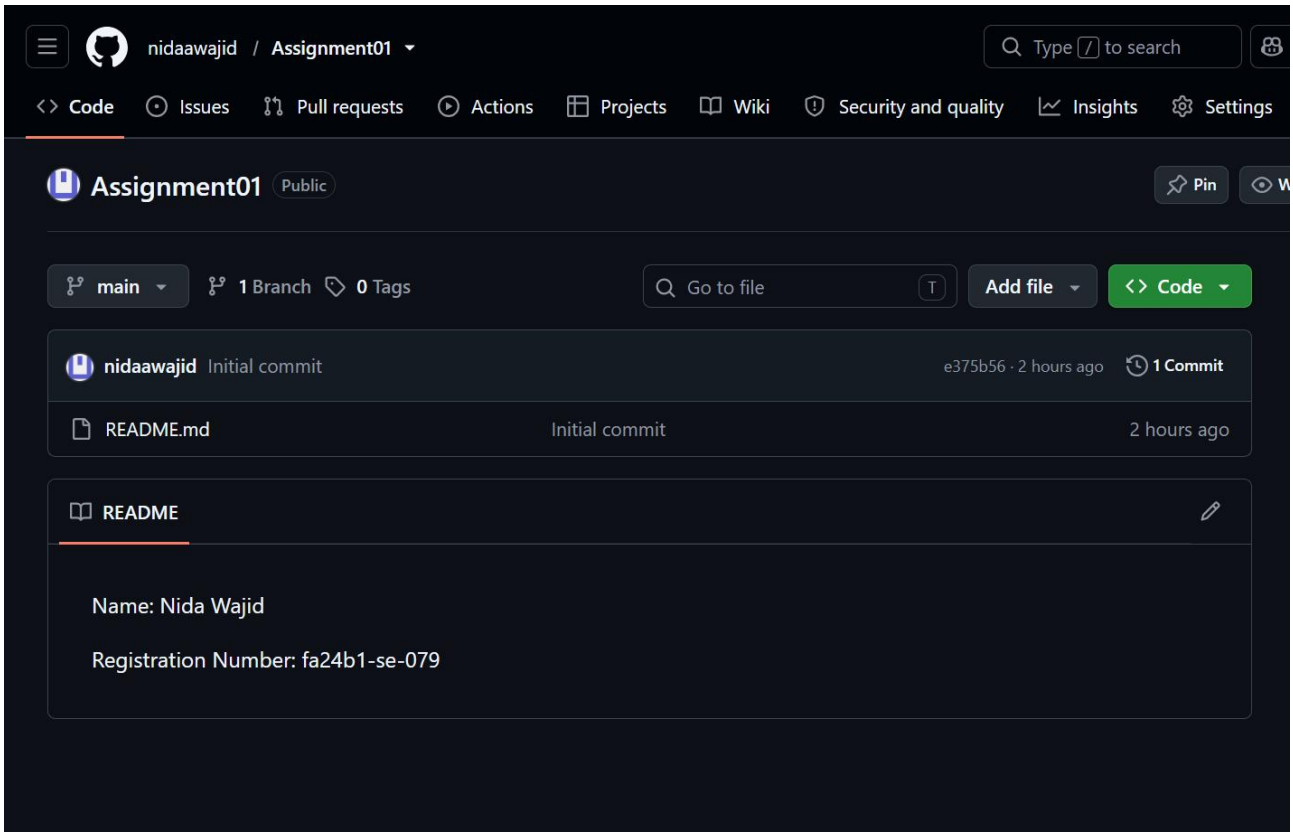
Step 2.2: Push to GitHub



Screenshot Name: `task2_github_push.png`

Figure 5: Main branch pushed successfully to the GitHub remote from the Ubuntu server.

Step 2.3: Verify the GitHub Repository



Screenshot Name: `task2_github_repository.png`

Figure 6: Public Assignment01 repository on GitHub with the rendered README showing name and registration number.

> **Task 2 result:** *The same local repository is connected to both Gitea and GitHub and is publicly available on GitHub.*

Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

```
nidaawajid@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
nidaawajid@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
nidaawajid@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
nidaawajid@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
nidaawajid@ubuntu:~/Assignment01$
```

Screenshot Name: `task3_lfs_setup.png`

*Figure 7: Git LFS initialized, its version confirmed, and the *.bin tracking rule shown in .gitattributes.*

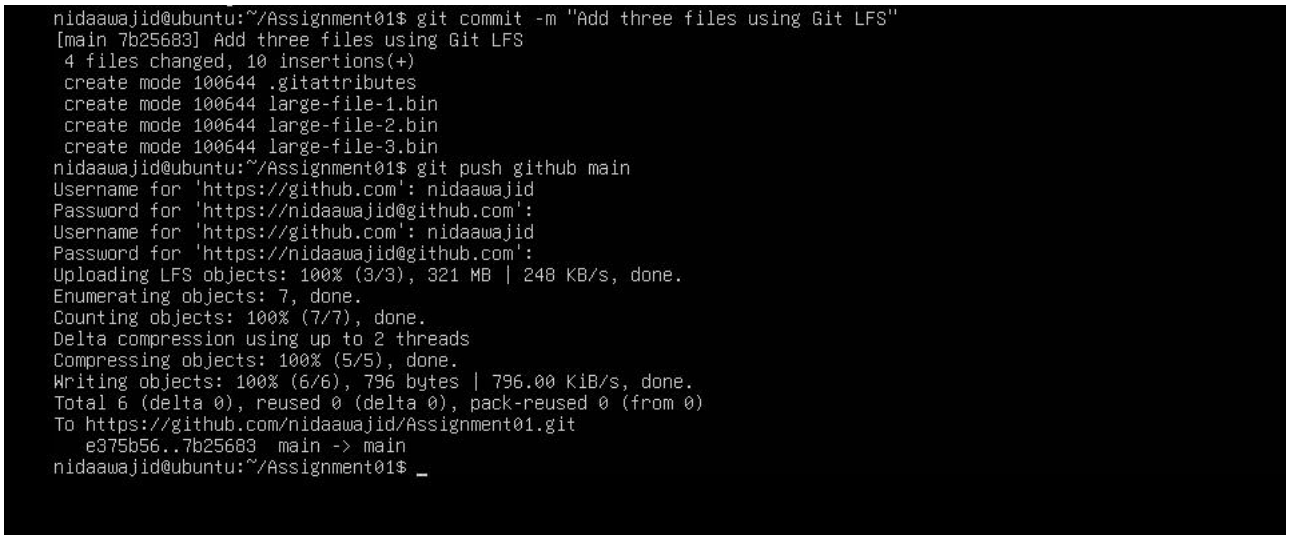
Step 3.2: Create and Verify Three LFS Files

```
nidaawajid@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
nidaawajid@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
nidaawajid@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
nidaawajid@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
nidaawajid@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
nidaawajid@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
nidaawajid@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
nidaawajid@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
nidaawajid@ubuntu:~/Assignment01$
```

Screenshot Name: `task3_lfs_files.png`

Figure 8: Three .bin files, each larger than 100 MiB, listed as tracked by Git LFS with their sizes.

Step 3.3: Commit and Push the LFS Files



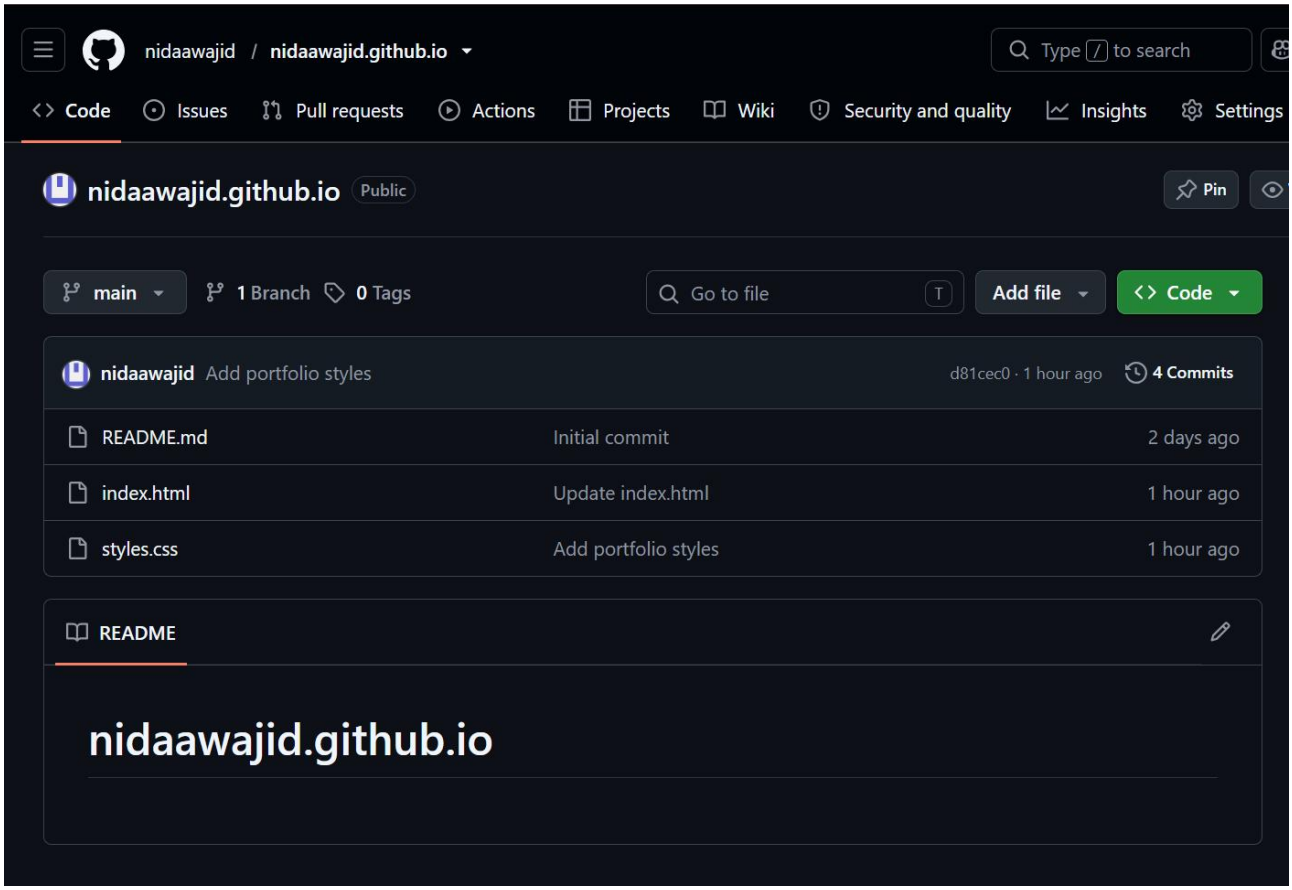
Screenshot Name: `task3_lfs_push.png`

Figure 9: Git LFS objects uploaded to GitHub successfully (3/3) along with the commit push.

> **Task 3 result:** *Three files larger than 100 MiB are tracked with Git LFS and pushed to the GitHub Assignment01 repository.*

Task 4: Create a Portfolio or CV with GitHub Pages

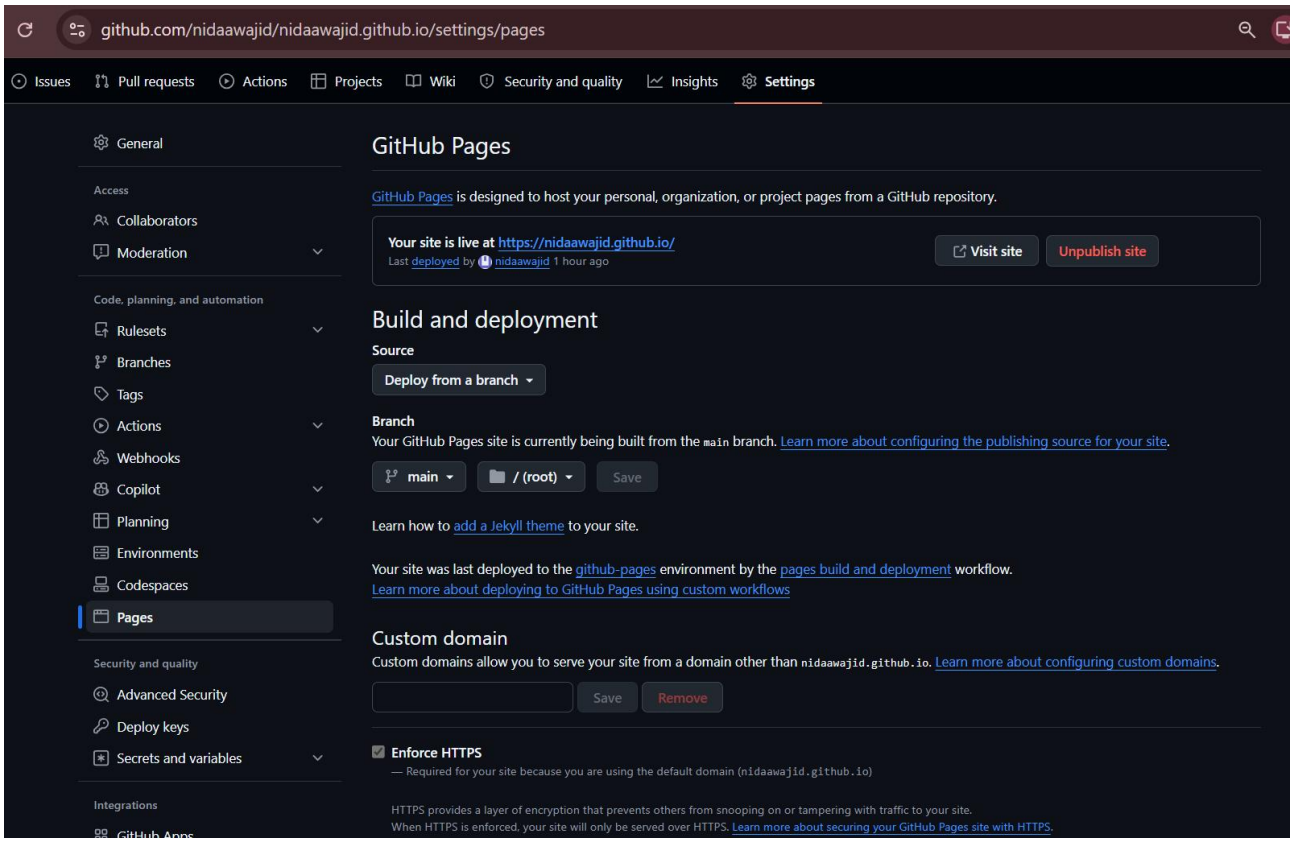
Step 4.1: Create and Push the Portfolio Repository



Screenshot Name: `task4_pages_repository.png`

Figure 10: Public nidaawajid.github.io repository showing the owner, repository name, index.html, and styles.css.

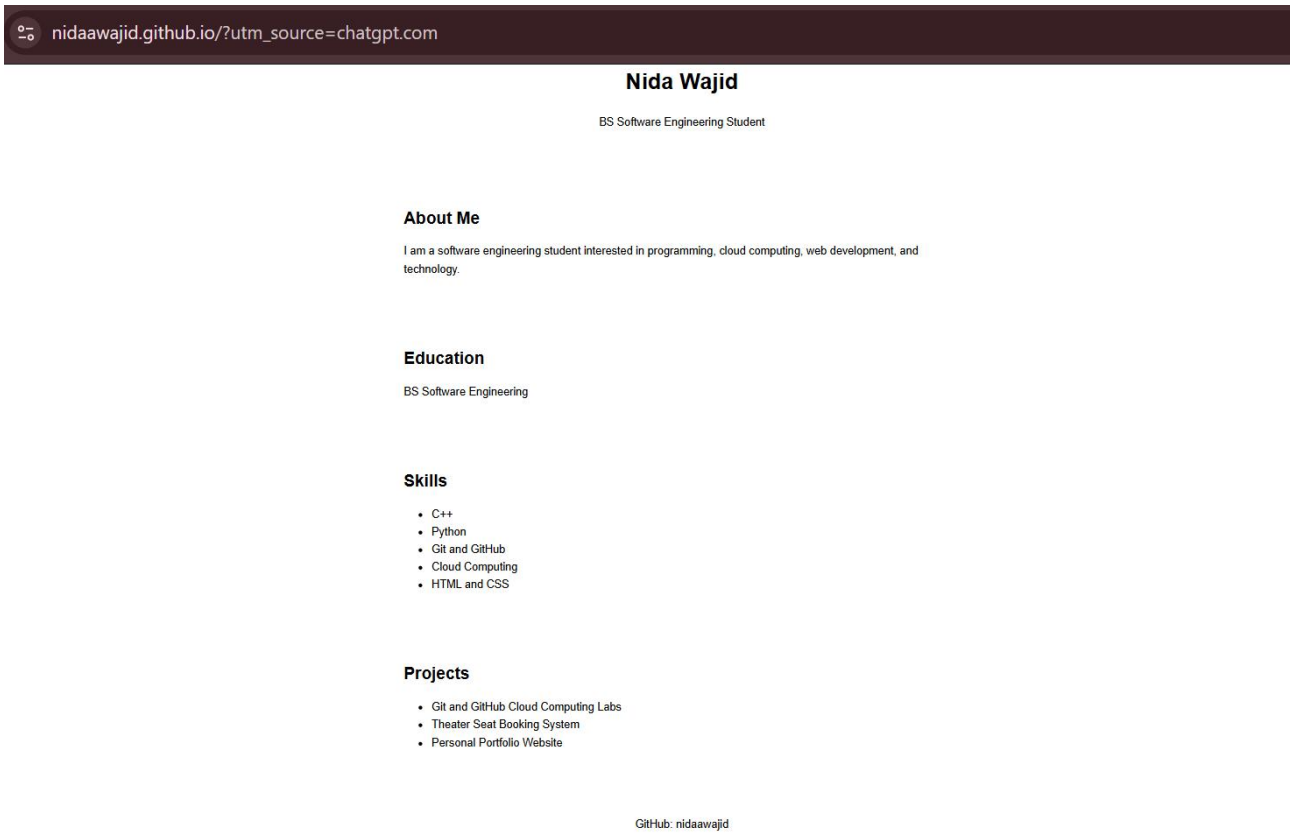
Step 4.2: Publish with GitHub Pages



Screenshot Name: `task4_pages_deployment.png`

Figure 11: GitHub Pages settings showing the site published from the main branch (root) with a live status.

Step 4.3: Verify the Live Portfolio



Screenshot Name: `task4_portfolio_live.png`

Figure 12: Live portfolio at https://nidaawajid.github.io/ displaying the required portfolio sections.

> **Task 4 result:** *The portfolio is stored in a public GitHub repository and published as a live GitHub Pages site.*